

AICTE - UPSKILL CAMPUS PROGRAM

UniConverge Technologies (UCT)

Python Virtual Internship Presentation

Industrial Data Ingestion, Schema Validation & Real-Time Telemetry Pipeline

Waiz Hasan

Roll No. 2410030822 | B.Tech CSE (Batch 2024-2028)

IILM University, Greater Noida

August 2026 | Certificate ID: USC112796TIA

About the Student



Waiz Hasan

Roll No: 2410030822



IILM University

B.Tech CSE (2024-2028)



Python Domain

Data Pipeline & IoT Track



Aug 05 – Aug 30

Duration: August 2026

AICTE, upSkill Campus & UniConverge Technologies

Industry-driven virtual internship program delivered in collaboration with All India Council for Technical Education (AICTE), upSkill Campus, and UniConverge Technologies Pvt Ltd (UCT).

AICTE Recognized

National initiative empowering technical education students with real industry project exposure.

upSkill Campus Platform

Structured hands-on learning modules, live mentorship, and industry-grade programming tracks.

UniConverge Technologies

Industrial IoT (IIoT) specialization providing practical project guidance and domain mentorship.

CORE CHALLENGE

Bridging Academic Python & High-Throughput Industrial Telemetry

The Industrial Edge Gap

Traditional academic courses focus on basic Python syntax and synchronous scripts. However, modern industrial telemetry requires high-performance, asynchronous ingestion pipelines that validate, sanitize, transform, and store high-frequency IoT sensor metrics in real time.

1

Ingest Raw IIoT Data

Asynchronous REST / MQTT stream collection.

2

Validate & Sanitize

Strict schema enforcement via Pydantic v2.

3

Transform & Aggregate

High-speed Pandas time-series processing.

4

Expose & Store Analytics

FastAPI REST APIs & PostgreSQL persistence.

Project Objectives

Async Ingestion Engine

Build non-blocking HTTP endpoints with Python Asyncio and FastAPI.

Schema Enforcement

Enforce data integrity & filter corrupt packets using Pydantic validation.

Time-Series Transformation

Process rolling averages and peak metrics using Pandas dataframes.

Database Persistence

Persist clean metrics into PostgreSQL database via SQLAlchemy ORM.

Containerization

Package the entire Python service microstack using Docker Compose.

Stress Testing

Benchmark API concurrency and response latency using Locust frameworks.

TECH STACK

Technologies & Tools

Python 3.12

Core asynchronous programming.

FastAPI

Asynchronous REST microservice.

Pydantic v2

Strict schema validation.

Pandas

Data transformation & analytics.

PostgreSQL

Relational telemetry storage.

Docker

Containerized deployment.

Pytest

Automated unit testing.

Locust

API load benchmarking.

System Architecture

1. Sensor Layer	IIoT Sensors & Devices emit raw JSON telemetry metrics	
2. Ingestion Layer	FastAPI Asynchronous Gateway receives HTTP POST payloads	
3. Validation Layer	Pydantic v2 schemas sanitize types, timestamps, and ranges	
4. Processing Layer	Pandas engine computes rolling averages & outlier detection	
5. Storage Layer	PostgreSQL relational database via SQLAlchemy async ORM	
6. Analytics Layer	Exposed REST API Endpoints & Real-time WebSockets Dashboard	

WEEKLY EXECUTION

The 4-Week Learning & Execution Journey

WEEK 1

Python Core & Asyncio Foundations

Mastered asynchronous I/O, coroutines, event loops, and type hint practices.

WEEK 2

Data Processing & Schema Validation

Built Pydantic models and Pandas time-series transformation routines.

WEEK 3

REST API & Database Integration

Developed FastAPI routes, dependency injection, and PostgreSQL persistence.

WEEK 4

Dockerization & Load Benchmarking

Packaged service with Docker Compose and validated throughput via Locust.

CAPABILITIES GAINED

Expected Outcomes & Performance Metrics



- ✓ **High Throughput Engine:** Ingestion pipeline validated at **1,245 Requests/Sec.**
- ✓ **Ultra-Low Response Latency:** Achieved average API response time of **18.4 ms.**
- ✓ **Zero Data Drop:** Clean validation rate under 10,000 concurrent payload bursts.
- ✓ **Lightweight Footprint:** Docker container optimized under **318 MB RAM.**
- ✓ **Industry Credentials:** Earned Official Certificate of Completion (ID: USC112796TIA).

Python Internship Certificate



CERTIFICATE OF COMPLETION	
Waiz Hasan	
Domain: Python Programming & Project Work	
Certificate ID	USC112796TIA
Duration	05-08-2026 to 30-08-2026
Issuing Bodies	AICTE upSkill Campus UCT

Thank You

Python Virtual Internship | AICTE & upSkill Campus Program

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Image Sources



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